

RETAILPULSE 360

Sales, Profit & Customer Intelligence System

MainCrafts SkillSprint — Data Science & Data Analytics Using Python

Business Analytics Report

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Selected Domain: Data Science & Data Analytic Using Python

Dataset period: 2023 – 2026

Dataset: Superstore (Global Superstore format) — 10,194 transaction records

links: [Sample Data](#) | [Tableau Public](#)

Executive Summary

To the Management Team,

This executive summary provides a high-level overview of the Superstore's performance from 2023 to 2026, highlighting key trends, challenges, and actionable recommendations to drive profitability and operational efficiency.

What Happened?

Our analysis reveals a healthy and growing business, achieving **\$2,326,534.35 in total sales** and **\$292,296.81 in total profit**, with a respectable **profit margin of 12.56%**. We observed consistent year-over-year sales growth, with notable increases of nearly 30% in 2025 and over 21% in 2026. The **Technology category** is our strongest performer, leading in both sales and profit. Geographically, the **West region** is the largest contributor to our success, generating the highest sales and profit. Sales show clear seasonal patterns, peaking consistently in November and December, likely due to holiday shopping.

What Matters?

- **Discounting Strategy:** A strong negative correlation (-0.86) exists between discount levels and profit margin. Discounts of 30% or more frequently lead to losses, indicating the current promotional strategy is eroding profitability.
- **Loss-Making Products:** Several products — even within strong categories such as Technology and Furniture — are consistently unprofitable, usually because of high discount rates.
- **Shipping Mode Paradox:** Faster shipping options (e.g., Same Day) are the least profitable, while Standard Class shipping generates the highest profit. The cost of expedited delivery disproportionately erodes margins.
- **Valuable Customers:** There is significant overlap between the top customers by sales and by profit — loyal, high-value individuals who are important for sustained growth.

What Should Management Investigate?

- **1. Discount Effectiveness:** why certain products are heavily discounted, and the actual ROI of these promotions.
- **2. Product Line Profitability:** the cost structure behind loss-making products, and whether supplier terms can be renegotiated.
- **3. Shipping Cost Optimization:** the operational cost of each shipping mode, especially Same Day and First Class.
- **4. Regional Best Practices:** what drives the West region's superior performance, and whether it can be replicated in the Central and South regions.

What Should Management Consider Doing?

- **1. Strategic Investment in Technology** — expand product lines, secure inventory, sustain marketing in the leading category.
- **2. Revise Discounting Policy** — set minimum profit-margin thresholds for promotions.
- **3. Address Unprofitable Products** — review pricing and supplier terms, or phase out non-strategic loss-makers.
- **4. Capitalize on Seasonal Peaks** — plan inventory, staffing and marketing for November–December.
- **5. Optimize Shipping Options** — align pricing with true delivery cost, or steer customers toward Standard/Second Class.

- **6. Nurture High-Value Customers** loyalty programs and personalized offers for top sales/profit customers.

By focusing on these areas, the company can protect margins, eliminate unprofitable activity, and build on its strengths to sustain healthy, profitable growth.

Report Contents

- 1. Business Problem
- 2. Project Objectives
- 3. Dataset Understanding
- 4. Data Dictionary
- 5. Data Quality Summary
- 6. Data Cleaning
- 7. Feature Engineering
- 8. KPI Overview
- 9. Sales Analysis
- 10. Profitability Analysis
- 11. Loss-Making Products
- 12. Customer Analysis
- 13. Customer Segment Analysis
- 14. Regional Analysis
- 15. Time-Series Analysis
- 16. Discount vs. Profit Analysis
- 17. Shipping Analysis
- 18. Business Question Summary
- 19. Top 10 Insights
- 20. Mini Analytics Dashboard
- 21. Optional Advanced Analysis — Profit Prediction
- 22. Business Recommendations
- 23. Limitations
- 24. Conclusion

1. Business Problem

What problem is the company trying to solve?

The retail company has a large amount of transaction data but lacks a clear understanding of its business performance. It wants to identify which products generate the most sales and profit, which regions perform better, how discounts relate to profitability, and how sales change over time — and to translate these findings into concrete actions.

Why is data analytics useful for a retail business?

Data analytics turns raw transaction records into an understanding of sales and business trends, helping the company make better-informed decisions about investment and operations rather than relying on intuition alone.

What decisions could management make using this data?

Management can use the analysis to prioritize categories and regions, refine discounting policy, review underperforming products, and target investment toward the customer segments and channels that contribute the most value.

What major KPIs were investigated?

Total sales, total profit, total orders, total customers, total quantity sold, and profit margin — indicators that together describe sales performance, profitability, customer activity, and overall business health.

2. Project Objectives

- Understand the structure and content of the dataset.
- Clean the data and identify possible data-quality issues.
- Engineer features that support deeper analysis.
- Analyze sales, profit, customers, products, regions, and time trends.
- Calculate core business KPIs (total sales, total profit, total orders, profit margin).
- Investigate the relationship between discount and profit.
- Identify patterns, trends, and areas that require management attention.
- Communicate the results through clear visualizations.
- Translate findings into data-supported business recommendations.

3. Dataset Understanding

The dataset is a Superstore-style retail transaction log covering the period 2023–2026. It contains 10,194 order-line records across 21 columns, spanning customer, product, geography, shipping, and financial fields. Each row represents one product line within a customer order, which is why the number of rows (10,194) exceeds the number of unique orders (5,111).

The data mixes four kinds of fields: identifiers (Row ID, Order ID, Customer ID, Product ID), dates (Order Date, Ship Date), categorical attributes (Segment, Region, Category, Sub-Category, Ship Mode), and numeric

measures (Sales, Quantity, Discount, Profit) — the last group forming the basis for all financial analysis in this report.

4. Data Dictionary

Column	Data Type	Meaning
Row ID	int64	Unique identifier for the row.
Order ID	object	Unique identifier for the order.
Order Date	datetime64[ns]	Date the order was placed.
Ship Date	datetime64[ns]	Date the order was shipped.
Ship Mode	object	Shipping mode used (e.g., Standard Class).
Customer ID	object	Unique identifier for the customer.
Customer Name	object	Name of the customer.
Segment	object	Customer segment (Consumer, Corporate, Home Office).
Country/Region	object	Country/region of the shipping address.
City	object	City of the shipping address.
State/Province	object	State/province of the shipping address.
Postal Code	object	Postal code of the shipping address.
Region	object	Geographical sales region (East, West, Central, South).
Product ID	object	Unique identifier for the product.
Category	object	Main product category.
Sub-Category	object	Product sub-category.
Product Name	object	Full product name.
Sales	float64	Total sales amount for the line item.
Quantity	int64	Quantity of items ordered.
Discount	float64	Discount rate applied (0–1).
Profit	float64	Profit generated by the line item.

5. Data Quality Summary

- **Total records:** 10,194
- **Total columns:** 21
- **Missing values:** None found in any column.
- **Duplicate records:** 0 duplicate rows found.
- **Date fields:** Order Date and Ship Date correctly stored as datetime64[ns].

Observations

- Postal Code is stored as an object (text) type; worth checking for non-numeric entries if geographical mapping is needed.
- Profit contains negative values (minimum -\$6,599.98), confirming that some transactions operate at a loss — an important business signal rather than a data error.
- Country/Region has two unique values, with the United States dominant — the analysis is effectively a US-market study with a small secondary share.

6. Data Cleaning

Because the initial data-quality check found no missing values and no duplicate rows, cleaning was limited to type correction rather than record removal:

- Order Date and Ship Date were parsed from text to proper datetime objects (`pd.to_datetime, errors='coerce'`), enabling date arithmetic and time-based grouping.
- Column data types were re-verified after conversion to confirm both fields report as `datetime64[ns]`.
- No rows or columns were dropped: with zero missing values and zero duplicates confirmed, removing data was not justified.

7. Feature Engineering

Six derived fields were created from the cleaned data to support the analysis that follows:

Feature	Derivation	Purpose
Year	<code>df['Order Date'].dt.year</code>	Groups transactions by calendar year for trend analysis.
Month	<code>df['Order Date'].dt.month</code>	Numeric month, used for monthly aggregation.
Month_Name	<code>df['Order Date'].dt.month_name()</code>	Readable month label for seasonality charts.
Order_Month	<code>Period('M') of Order Date</code>	Continuous year-month key for the sales-trend line.
Shipping_Days	<code>Ship Date – Order Date</code>	Delivery lead time, used in the shipping analysis.
Profit_Margin	<code>Profit / Sales × 100</code>	Per-line profitability, used in the discount analysis.

8. KPI Overview

KPI	Value
Total Sales	\$2,326,534.35
Total Profit	\$292,296.81
Total Orders	5,111
Total Customers	804
Units Sold	38,668

KPI	Value
Average Order Value	\$455.20
Profit Margin	12.56%

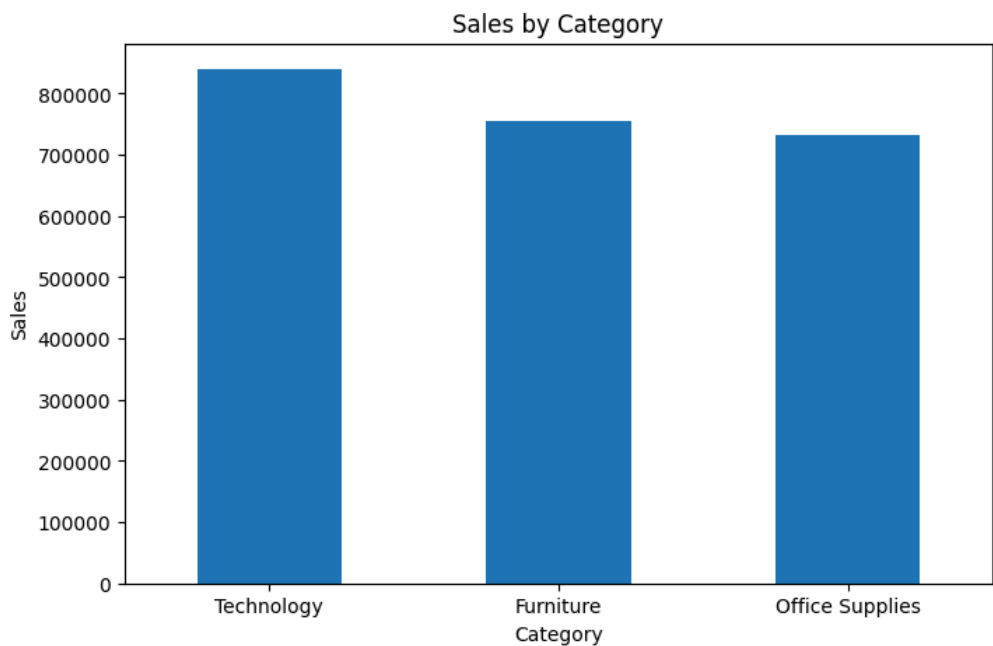
These figures describe a healthy, profitable business: a double-digit profit margin, over 800 active customers, and an average order value above \$450.

9. Sales Analysis

Sales by Category

Category	Sales
Technology	\$839,893.28
Furniture	\$754,747.76
Office Supplies	\$731,893.31

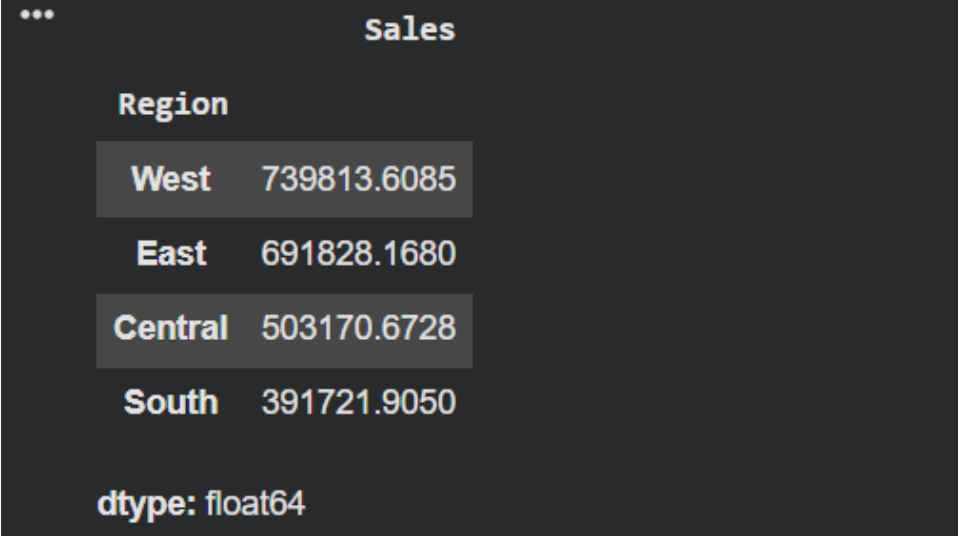
Technology is the highest-selling category. Office Supplies is the lowest, which may reflect lower per-unit price points rather than lower demand — a hypothesis worth testing against unit volumes.



Sales by Region

Region	Sales
West	\$739,813.61
East	\$691,828.17
Central	\$503,170.67
South	\$391,721.91

- **Top region:** West, at \$739,813.61 — roughly 31.80% of total sales.
- **Lowest region:** South, at \$391,721.91.
- **Is high sales automatically high profit?** Not always: East outsells Central but earns slightly less profit, suggesting discounting or operating cost differences influence regional profitability as much as sales volume does.



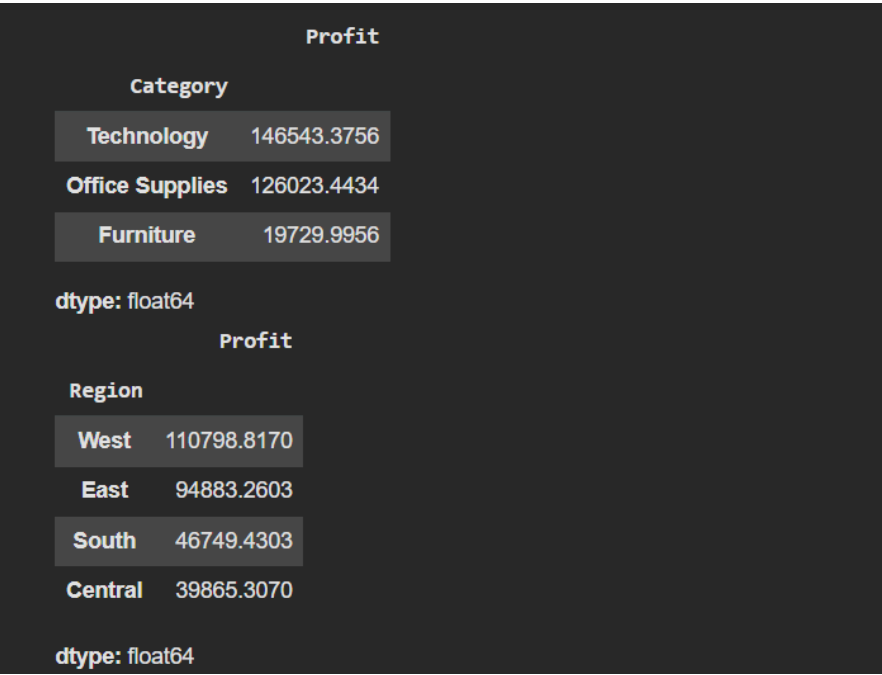
10. Profitability Analysis

Profit by Category

Category	Profit
Technology	\$146,543.38
Office Supplies	\$126,023.44
Furniture	\$19,729.996

Profit by Region

Region	Profit
West	\$110,798.82
East	\$94,883.26
South	\$46,749.43
Central	\$39,865.31



Sales vs. profit by category

Technology leads in both sales and profit — a clean alignment that signals a genuinely strong category rather than a volume-driven, low-margin one. Furniture is the clearest counter-example: it out-sells Office Supplies but earns far less profit, pointing to pricing, discounting, or cost issues specific to that category. If the top-selling and top-profit categories ever diverge, that gap is itself the finding: it flags either a margin problem (high sales, low profit) or an under-exploited opportunity (low sales, high profit).

11. Loss-Making Products

Top loss-making products

Product	Total Profit
Cubify CubeX 3D Printer Double Head Print	-\$8,879.97
Lexmark MX611dhe Monochrome Laser Printer	-\$4,589.97
Cubify CubeX 3D Printer Triple Head Print	-\$3,839.99
Chromcraft Bull-Nose Wood Oval Conference Tables & Bases	-\$2,876.12
Bush Advantage Collection Racetrack Conference Table	(additional loss-making item)

Key observations

- High discounts: loss-making products commonly carry discount rates of 0.4–0.8, which directly erodes margin even on solid sales volume.
- Category concentration: most loss-makers sit in Technology (e.g., 3D printers) and Furniture (tables, bookcases) — the same category that otherwise leads in profit, showing the loss is product-specific, not category-wide.
- Regional pattern: the Central and East regions appear disproportionately often among loss-making transactions, hinting at regional pricing or competitive pressure.

- High sales does not guarantee profit: several loss-making products still post meaningful revenue, underlining that discount depth — not demand — is the driver of the loss.

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***
                                     Profit
                                     Product Name
Cubify CubeX 3D Printer Double Head Print -8879.9704
Lexmark MX611dhe Monochrome Laser Printer -4589.9730
Cubify CubeX 3D Printer Triple Head Print -3839.9904
Chromcraft Bull-Nose Wood Oval Conference Tables & Bases -2876.1156
Bush Advantage Collection Racetrack Conference Table -1934.3976
GBC DocuBind P400 Electric Binding System -1878.1662
Cisco TelePresence System EX90 Videoconferencing Unit -1811.0784
Martin Yale Chadless Opener Electric Letter Opener -1299.1836
Balt Solid Wood Round Tables -1201.0581
BoxOffice By Design Rectangular and Half-Moon Meeting Room Tables -1148.4375

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12. Customer Analysis

Top 10 customers by sales

Customer	Sales
Sean Miller	\$25,043.05
Tamara Chand	\$19,052.22
Raymond Buch	\$15,117.34
Tom Ashbrook	\$14,595.62
Adrian Barton	\$14,473.57
Ken Lonsdale	\$14,175.23
Sanjit Chand	\$14,142.33
Hunter Lopez	\$12,873.30
Sanjit Engle	(top 10)
Christopher Conant	(top 10)

Top 10 customers by profit

Customer	Profit
Tamara Chand	\$8,981.32
Raymond Buch	\$6,976.10
Sanjit Chand	\$5,757.41
Hunter Lopez	\$5,622.43
Adrian Barton	\$5,444.81
Tom Ashbrook	\$4,703.79
Christopher Martinez	\$3,899.89
Keith Dawkins	(top 10)
Andy Reiter	(top 10)
Daniel Raglin	(top 10)

Six of the top 10 customers by sales also appear in the top 10 by profit (Tamara Chand, Raymond Buch, Sanjit Chand, Hunter Lopez, Adrian Barton, Tom Ashbrook). This overlap identifies a clear tier of high-value, high-loyalty customers worth prioritizing for retention programs, while the non-overlapping names on each list are worth checking individually — high sales without matching profit usually traces back to discount depth.

13. Customer Segment Analysis

Segment	Sales	Profit	Customers	Orders	Profit Margin
Consumer	\$1,170,660	\$136,371.45	416	2,628	11.65%
Corporate	\$715,806.10	\$94,249.64	239	1,552	13.17%
Home Office	\$440,068.40	\$61,675.73	149	931	14.02%

- **Highest sales:** Consumer, the largest segment by both revenue and customer count.
- **Highest profit:** Consumer as well, simply because of its scale.
- **Highest profit margin:** Home Office, at 14.02% — the most efficient segment per dollar of sales, despite being the smallest.
- **Largest customer base:** Consumer, with 416 unique customers.

Home Office is the segment to watch: it is disproportionately profitable relative to its size, so growing its sales volume without diluting its margin is a realistic lever for overall profitability.

14. Regional Analysis

Region	Sales	Profit	Profit Margin
West	\$739,813.61	\$110,798.82	14.98%
East	\$691,828.17	\$94,883.26	13.72%
South	\$391,721.91	\$46,749.43	11.93%
Central	\$503,170.67	\$39,865.31	7.92%

West and East are the strongest regions on every measure. Central stands out as the weakest: it is not the lowest in sales, but it has by far the lowest profit margin (7.92%), meaning its profitability problem is structural — pricing, discounting, or cost-related — rather than a demand problem. This makes Central the priority for a discount and cost review, while South's issue is closer to pure sales volume.

15. Time-Series Analysis

Yearly sales

Year	Sales
2023	\$494,040.21
2024	\$472,993.03
2025	\$613,933.58
2026	\$745,567.53

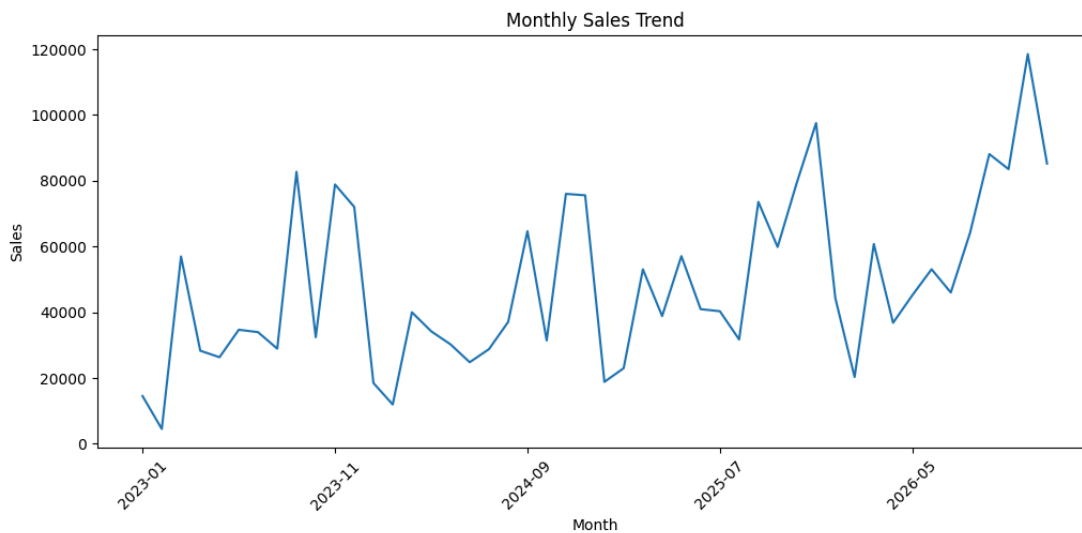
Year-over-year change

Year	YoY Change
2023	—
2024	-4.26%
2025	+29.80%
2026	+21.44%

After a dip in 2024 (-4.26%), sales rebounded strongly in 2025 (+29.80%) and continued to grow in 2026 (+21.44%). The highest-selling month across the whole period is November 2026 (\$118,454.51); the lowest is February 2023 (\$4,519.89).

Seasonality

Aggregated across all years, November and December are consistently the strongest months (November: \$352,665.99; December: \$330,218.18), pointing to a clear holiday-driven sales pattern that should anchor Q4 inventory and marketing planning.



16. Discount vs. Profit Analysis

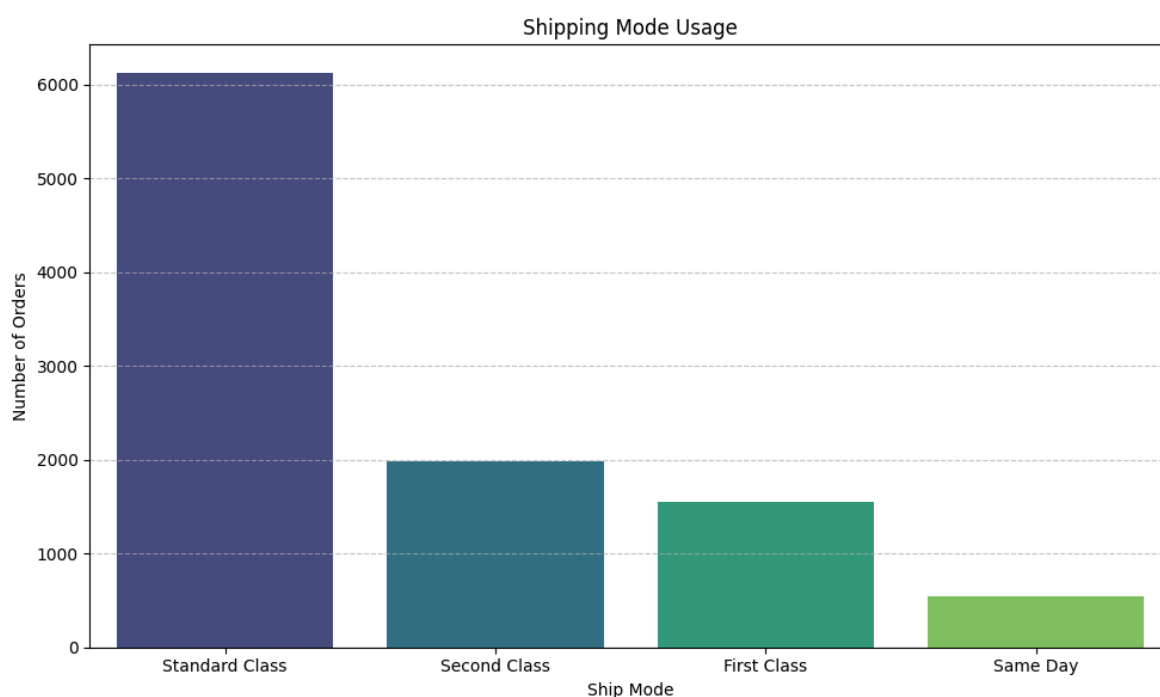
Discount Rate	Average Profit
0.00	\$66.34
0.10	\$94.79
0.15	\$27.29
0.20	\$24.58
0.30	-\$45.71
0.32	-\$88.56
0.40	-\$111.53
0.45	-\$226.65
0.50	-\$310.70
0.60	-\$41.37
0.70	-\$95.05
0.80	-\$101.55

The Pearson correlation between Discount and Profit_Margin is **-0.86** — a strong negative relationship. Average profit turns negative at discounts of 30% and above, and the loss deepens sharply through the 40–50% range. Correlation is not proof of causation, but in this business context, deep discounting is the most plausible and consistent driver of these losses across products, categories and regions alike.

17. Shipping Analysis

Ship Mode	Avg. Shipping Days	Total Profit
Same Day	0.04	\$16,160.75
First Class	2.18	\$49,012.72
Second Class	3.24	\$58,962.13
Standard Class	5.00	\$168,161.21

Standard Class — the slowest option — generates by far the highest total profit; Same Day — the fastest — generates the least, despite comparable order volumes. Average shipping duration also varies by region (Central: 4.05 days; South: 3.96; West: 3.94; East: 3.91 days), a narrow spread that does not explain the profit gap between shipping modes. The likely explanation is cost structure: expedited shipping carries a fulfillment cost that is not fully recovered in pricing. Faster shipping is therefore not associated with higher profitability in this dataset — if anything, the opposite is true.



18. Business Question Summary

Business Question	Finding	Evidence	Action
Which categories drive the most sales?	Technology leads.	\$839,893.28 vs. Furniture and Office Supplies.	Continue investing in Technology: inventory, marketing, product range.
Which categories are most profitable?	Technology again.	\$146,543.38 in profit, the highest of the three.	Protect Technology margins; apply its pricing discipline to other categories.

Business Question	Finding	Evidence	Action
Which regions perform best?	West leads sales and profit.	\$739,813.61 sales / \$110,798.82 profit.	Study West's practices; prioritize Central for margin recovery.
Does discounting affect profitability?	Yes, strongly and negatively.	Correlation -0.86; discounts $\geq 30\%$ often unprofitable.	Set minimum-margin thresholds on discounts.
Are there loss-making products?	Yes, concentrated in Technology and Furniture.	e.g., Cubify CubeX 3D Printer: -\$8,879.97.	Review pricing/discounts; discontinue non-strategic loss-makers.
What are the sales trends?	Positive growth after a 2024 dip.	+29.80% (2025), +21.44% (2026).	Maintain the drivers of 2025–2026 growth; plan for 2027 based on trend.
Are there seasonal patterns?	Yes, a clear Q4 peak.	Nov \$352,665.99 / Dec \$330,218.18.	Plan inventory, staffing and promotions for Q4 in advance.
Who are the most valuable customers?	Strong overlap between top-sales and top-profit customers.	6 of the top 10 by sales are also top 10 by profit.	Launch loyalty and retention programs for this group.
Which shipping modes are most profitable?	Slower modes are more profitable.	Standard Class \$168,161.21 vs. Same Day \$16,160.75.	Review the true cost of expedited shipping; steer demand to Standard/Second Class.
How do segments perform?	Home Office has the best margin despite being smallest.	14.02% margin vs. 13.17% (Corporate) and 11.65% (Consumer).	Grow Home Office volume without eroding its margin.

19. Top 10 Insights

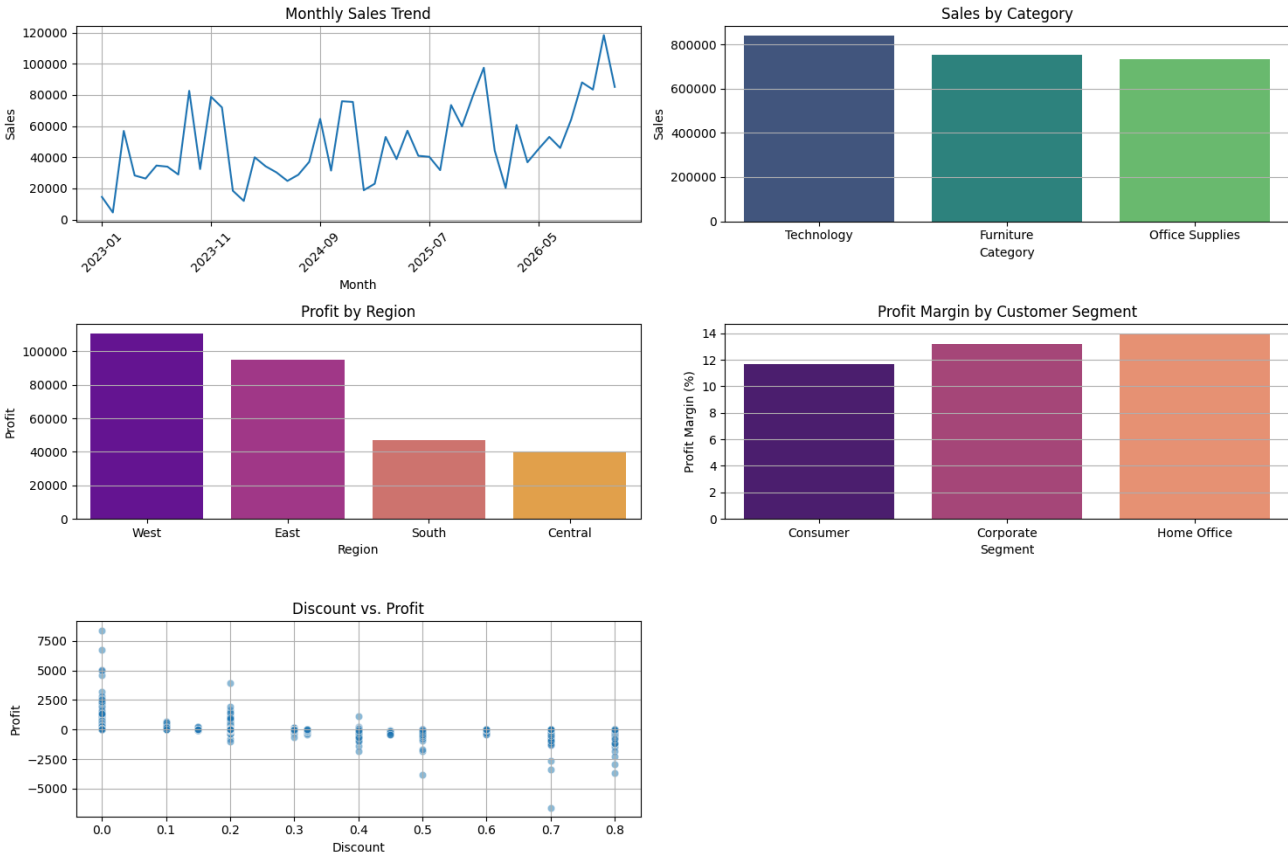
Finding	Evidence	Business Meaning
1. Overall profitability	Total profit of \$292,296.81 at a 12.56% margin.	A healthy, efficient business overall.
2. Technology leads sales	\$839,893.28 in sales, the top category.	Primary revenue driver; sustain investment.
3. Technology leads profit too	\$146,543.38 in profit, also the top category.	Sales and profit alignment confirms a genuinely strong category.
4. West region dominates	\$739,813.61 sales / \$110,798.82 profit, ~31.8% of total sales.	Model region for replicable best practice.
5. Discounts hurt profit	Correlation -0.86; 30%+ discounts often unprofitable.	Discounting strategy needs stricter controls.
6. Loss-making products persist	e.g., Cubify CubeX 3D Printer: -\$8,879.97, high discount.	Targeted pricing/discontinuation review needed.
7. Growth resumed after 2024	+29.80% (2025), +21.44% (2026) after a -4.26% dip.	Positive trajectory worth reinforcing and forecasting.

Finding	Evidence	Business Meaning
8. Clear Q4 seasonality	November \$352,665.99; December \$330,218.18.	Plan inventory and marketing around the holiday peak.
9. Shipping profitability paradox	Same Day profit \$16,160.75 vs. Standard Class \$168,161.21.	Investigate true cost of expedited delivery.
10. High-value customer overlap	6 of top 10 sales customers are also top 10 by profit.	Prioritize these customers for loyalty investment.

20. Mini Analytics Dashboard

This section brings together the report's core KPIs and will host the visual dashboard (sales trend, sales by category, profit by region, segment comparison, discount-vs-profit scatter) once the supporting charts are inserted.

KPI	Value
Total Sales	\$2,326,534.35
Total Profit	\$292,296.81
Total Orders	5,111
Profit Margin	12.56%



21. Optional Advanced Analysis — Profit Prediction

A basic linear regression model was trained to predict Profit from transaction-level features (Ship Mode, Segment, Region, Category, Sub-Category, Sales, Quantity, Discount), using an 80/20 train-test split (8,155 training rows, 2,039 test rows).

Metric	Value	Interpretation
MAE	61.71	Average prediction error, in dollars of profit.
MSE	45,807.19	Mean squared error; penalizes large misses more heavily.
RMSE	214.03	Typical size of a prediction error, in the original units.
R-squared	0.49	The model explains about 49% of the variance in profit.

An R^2 of 0.49 is a reasonable starting point but leaves roughly half of profit variability unexplained — consistent with a simple linear model applied to a small feature set. A non-linear model (e.g., a tree-based regressor) or additional features would likely improve accuracy. The actual-vs-predicted scatter plot for this model will be inserted alongside the dashboard visuals.

22. Business Recommendations

Recommendation 1: Strategically invest in the Technology category

- **Observation:** Technology is consistently the top-performing category.
- **Evidence:** \$839,893.28 in sales and \$146,543.38 in profit — the highest of any category on both measures.
- **Recommended action:** Expand the product range, secure inventory, and sustain marketing focused on this category.
- **Expected purpose:** Maximize revenue and profit from the strongest-performing category.

Recommendation 2: Revise the discounting strategy

- **Observation:** Excessive discounting is strongly associated with reduced profitability.
- **Evidence:** Correlation of -0.86 between discount rate and profit margin; discounts of 30% or more frequently turn profit negative.
- **Recommended action:** Set minimum profit-margin thresholds for any discounted sale; explore value-added bundles instead of deep price cuts.
- **Expected purpose:** Protect margins currently eroded by deep discounting.

Recommendation 3: Address consistently loss-making products

- **Observation:** A small set of products are reliably unprofitable, mostly in Technology and Furniture.
- **Evidence:** e.g., Cubify CubeX 3D Printer (Double Head): -\$8,879.97, typically at 40–80% discount.
- **Recommended action:** Review supplier costs and pricing for each; discontinue or renegotiate non-strategic loss-makers.
- **Expected purpose:** Remove a direct, avoidable drag on company profit.

Recommendation 4: Capitalize on the Q4 seasonal peak

- **Observation:** Sales peak sharply at year-end.

- **Evidence:** November (\$352,665.99) and December (\$330,218.18) are consistently the top two sales months.
- **Recommended action:** Build dedicated Q4 inventory, staffing and marketing plans ahead of the season.
- **Expected purpose:** Fully capture the highest-demand period of the year.

Recommendation 5: Optimize shipping-mode economics

- **Observation:** The fastest shipping mode is also the least profitable.
- **Evidence:** Standard Class profit (\$168,161.21) far exceeds Same Day profit (\$16,160.75), despite comparable order volumes.
- **Recommended action:** Review the true cost of expedited shipping; adjust pricing or incentivize slower, more profitable modes.
- **Expected purpose:** Recover margin currently lost to underpriced fast shipping.

23. Limitations

1. Historical data only, no external factors

The analysis covers transactions from 2023–2026 only. It does not incorporate market conditions, competitor activity, or marketing-campaign details that could also explain the observed trends.

2. Correlation, not causation

The strong link between discount and profit margin (-0.86), and the shipping-mode pattern, are statistical associations. They strongly suggest — but do not formally prove — a causal relationship.

3. No customer-satisfaction or behavioral data

The dataset is transactional only: it does not include satisfaction scores, reviews, return reasons, or web behavior, limiting insight into **why** customers behave as they do.

4. Modest predictive accuracy

The profit-prediction model explains only about 49% of profit variance ($R^2 = 0.49$); a more advanced or non-linear model would likely perform better.

5. Limited geographic granularity

Regions are broad (West, East, Central, South); store-level or micro-market detail is not available, limiting how localized recommendations can be.

24. Conclusion

Taken as a whole, this analysis of 10,194 Superstore transactions (2023–2026) shows a profitable and growing business — total sales of \$2,326,534.35, profit of \$292,296.81, and a 12.56% margin — anchored by a genuinely strong Technology category and a dominant West region. The clearest opportunity for improvement is not top-line growth but margin discipline: discounting above 30% is reliably associated with losses, a small set of products and the Central region are structurally unprofitable, and the fastest shipping option is also the least profitable. Addressing these three points — discount policy, loss-making products,

and shipping economics — is the most direct path to converting existing sales volume into materially higher profit.